PROJECT N°1

Schedule App



ABOUT ME

Johanne Vigouroux

UX/UI Designer with a Software Engineering Mindset

Engineering student turned designer. I sketch on paper, test with real students, then build the front-end myself.

- **Available for internships**

How I work

The process I followed to redesign IMT Atlantique's schedule app for students on their phones.

1 Research

Getting to know students first

Survey with 32 students
Review of the old app
Empathy maps
Personas

2 Plan

Turning needs into a clear structure

Scenario: next class in <10 s
Paper sketches
User flows
Information architecture
Wireframes

3 Make

Designing it, then building it

High-fidelity Figma prototypes
Micro-interactions
Accessible colors
Color-blind mode
Front-end built with Ionic

4 Test

Checking it with real students

Guerrilla tests with 20 people
Usability score (SUS)
Changes after each round of tests

ABOUT THE PROJECT

Next class in under 10 seconds

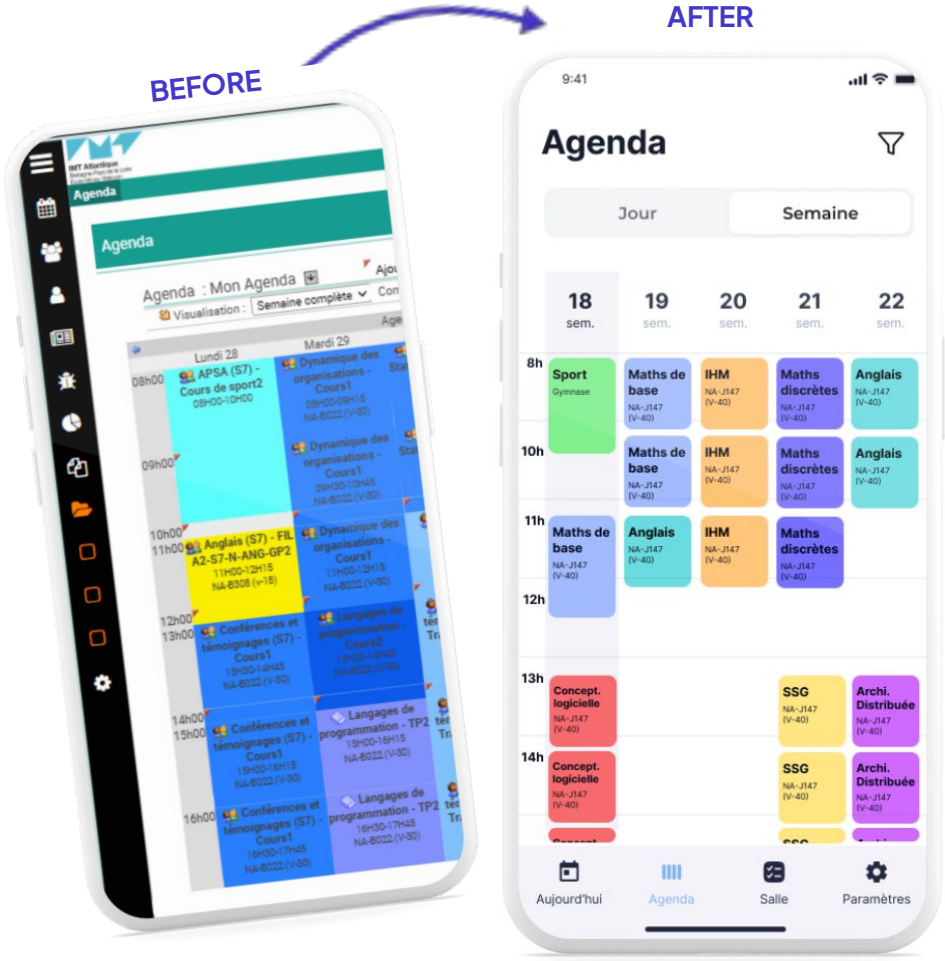
We redesigned IMT Atlantique's schedule app so students find their next class on their phone in seconds.

TIMELINE
1 weeks, 2026

TEAM
2 students

MY ROLE
Research, UI design, front-end

TOOLS
Figma, Ionic, Claude, Vercel, Notion



WHO IT'S FOR

Work-study students checking their schedule between classes.

THE PROBLEM

Built for desktop,
yet **22 of 32 students** use it on their phone.

WHAT WE BUILT

Opens on your next class.
Usability score **71/100** (**32 testers**).

THE PROBLEM

A desktop schedule squeezed onto a phone

What we found in the old app

Not made for phones

1. The icon-only side menu takes about a sixth of the width.
2. Desktop controls, like the week dropdown, are too small to tap.

Hard to read

3. Times and room codes are tiny, hard to read while walking.
4. Dark blue text on bright blue blocks: low contrast.
5. Hours are unevenly spaced, so start times are unclear.

Slow when every second of the break counts

6. About **3 s** to load on a phone. Source: **5 timed test / 7 survey**.
7. Nothing highlights today or the next class.



WHAT THIS MEANS FOR STUDENTS

An interface that generates daily stress

- **Arriving late :** A slow load between classes means missing the start of the lesson.
- **Wrong room :** A tiny room number can send students to the wrong building.
- **Constant double-checking :** Unclear names and colors make students re-check and still feel unsure.

22 out of **32**

students check their schedule on their phone.

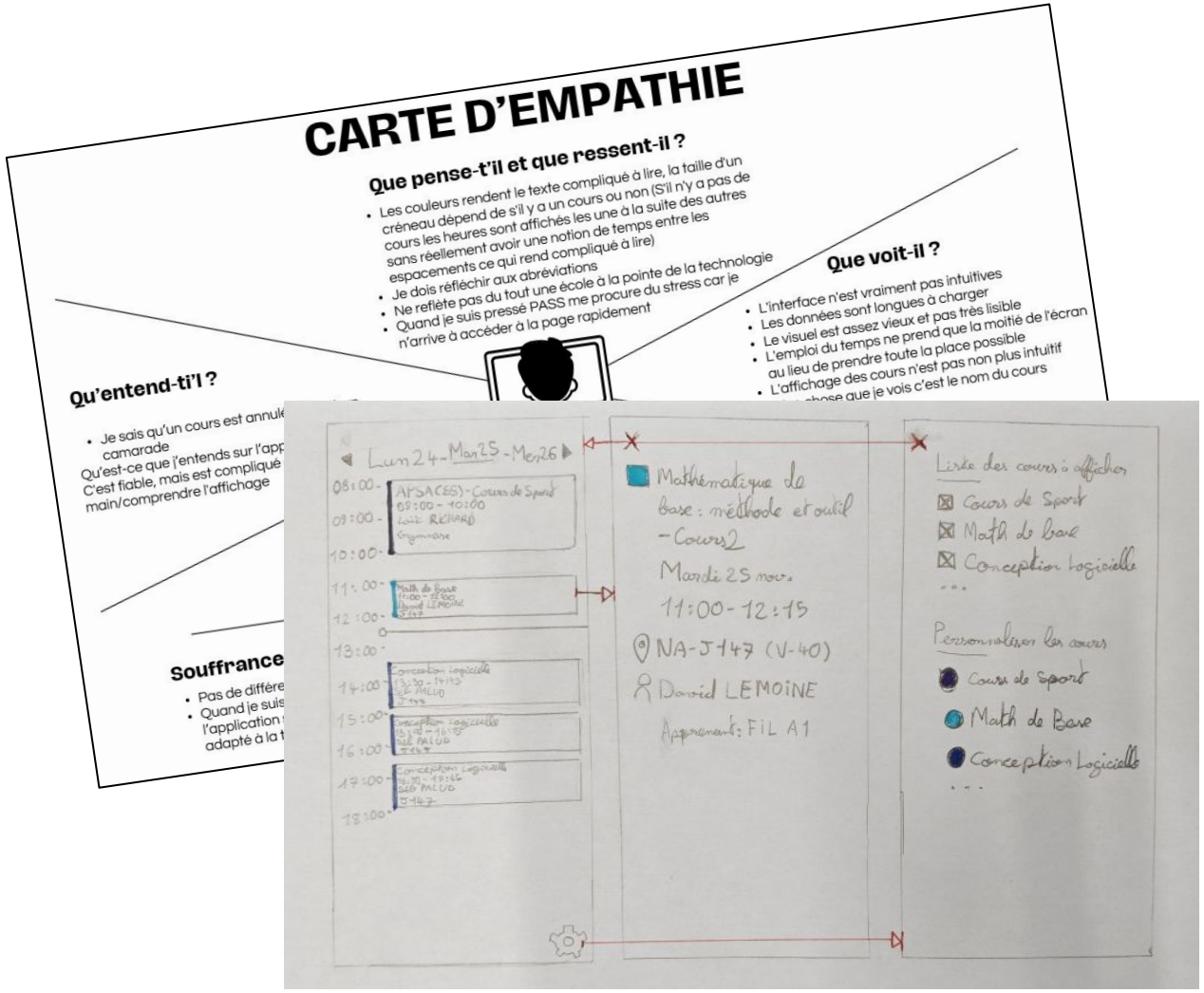
What students told us

From **12** student interviews

Hard on the eyes
The bright colors make the schedule hard to read.

Speed
I want to see my next class right away.

Personal colors
I want to spot each subject by its color.



From research to first sketch: part of the empathy map, a student persona and the paper wireframe of the main flow.

The moment we designed for

As a busy work-study student, I want to open the app and see my next class right away (time, room, subject), so I know where to go without wasting my break.

STEP 1

Between two classes

A student opens the app on their phone, already signed in with the school login.

STEP 2

No tapping needed

The app opens on today, with the next class visible at the top.

STEP 3

One tap for details

Tapping the class shows the teacher, the room and the type of class.

GOAL

Done in under 10 seconds

The student knows where to go before the break is over.

INTERFACE DESIGN

Calm, readable and made for phones

Components that fit small screens

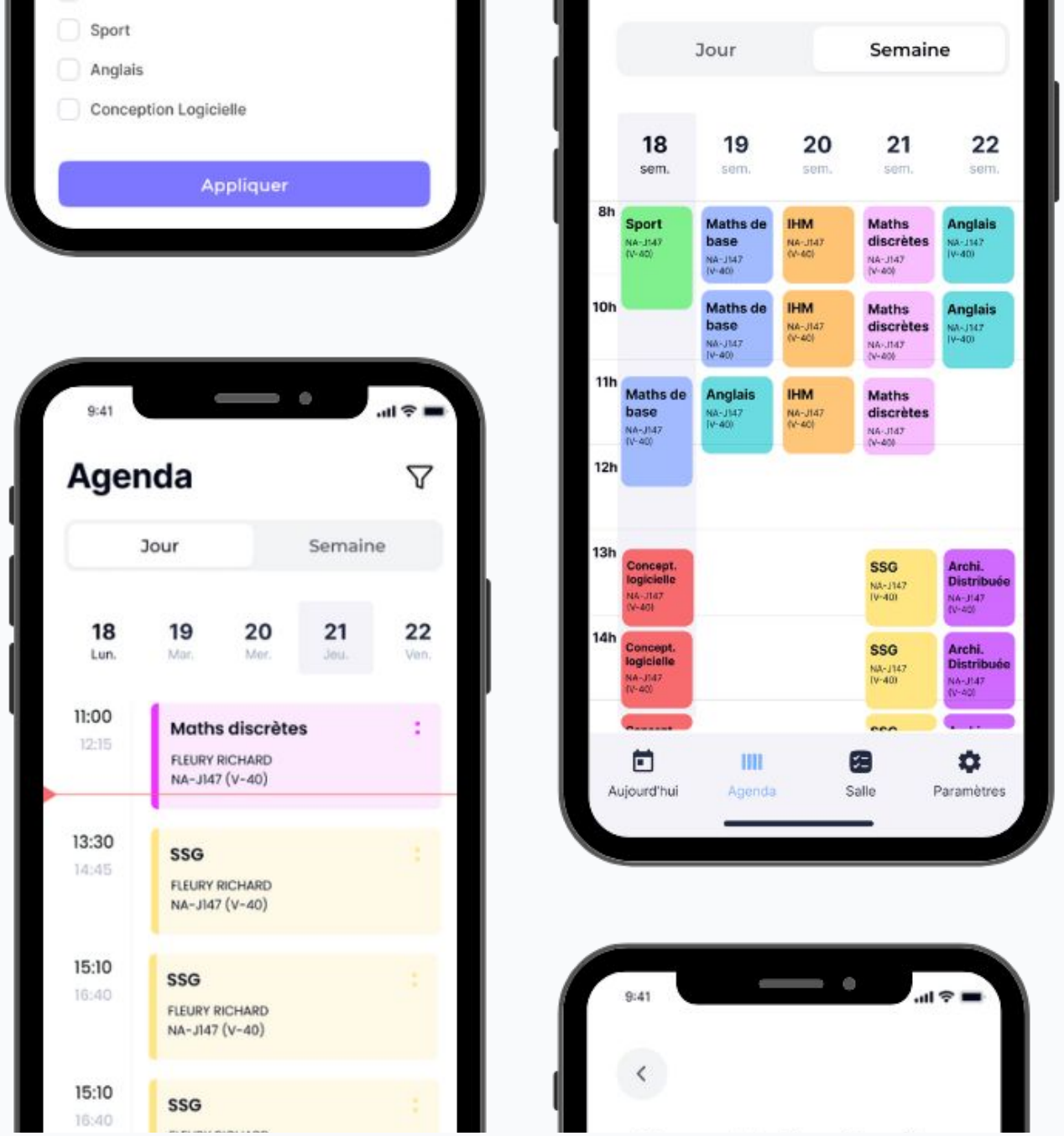
Built with Ionic, a toolkit of ready-made mobile components, so every screen behaves the same way.

Text that reads well on a phone

A clear type scale: class name first, then teacher and room.

Softer course colors

Pastel blocks with dark text replace the bright, tiring colors of the old app.



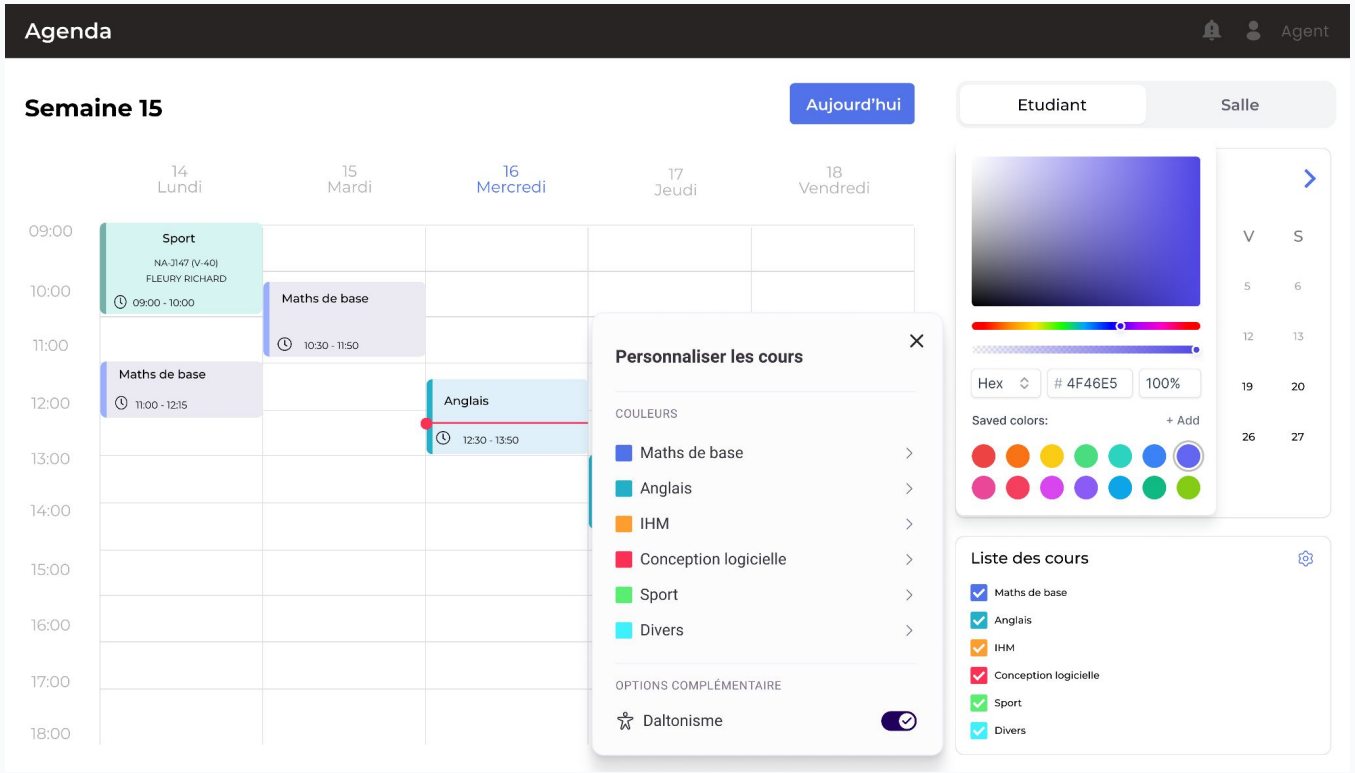
Day view, week view and course filters in the final prototype.

INCLUSIVITY FOCUS

Accessibility is not an option

Integration of a specific mode for color blindness and customization of colors

- Toggle color-blindness
- Customizable color code



While students check their phones on the go between classes to find a room, they rely on laptops when studying, doing coursework or planning their week ahead.

What testing changed

20 people tried the prototype

WHAT WE WANTED TO LEARN

- Can students find their next class in under 10 s?
- Is the navigation easy to follow?
- Is the schedule easy to read?

Class details were hard to tell apart

Users couldn't effortlessly distinguish the information from the schedule, so the colors of the information were changed to have a stronger contrast

We changed: darker text on the class cards, from 1.4:1 to 7.6:1 contrast.

Some navigation paths didn't work

The back arrow didn't take me where I expected, I felt lost.

We changed: made calendar headers clickable and added clear exit breadcrumbs

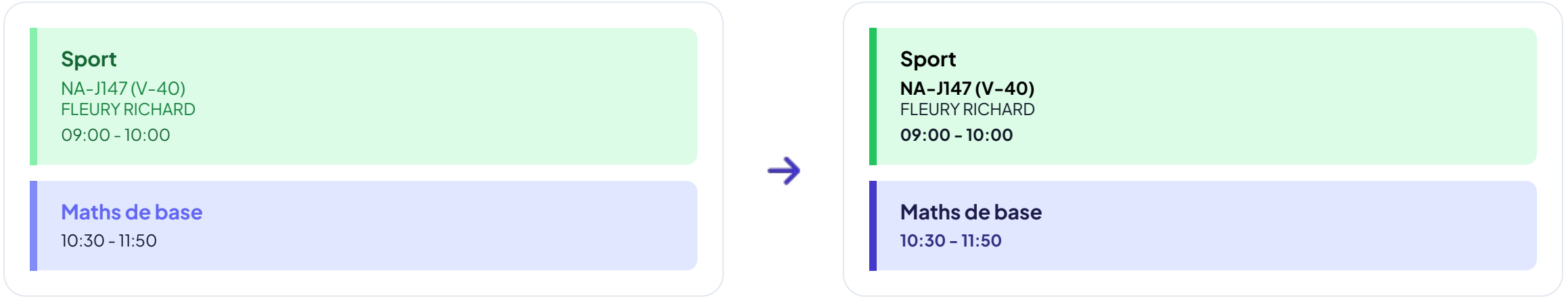
Students wanted to save notes on a class

It would be so useful to jot down reminders like "bring laptop" for specific classes.

We changed: added personal notes to each class.

Fixing contrast after testing

Testers struggled to read the class details, so we darkened the text on every card.

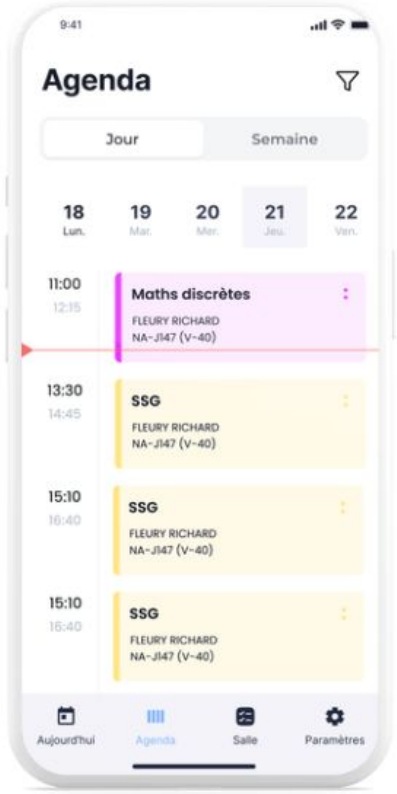


Version 1 — Sport card **1.4:1** · Maths **3.8:1**

Version 2 — **Sport card 7.6:1 · Maths 6.4:1**

Contrast ratios measured on the card text. The minimum for readable body text is 4.5:1.

From opening the app to the right room



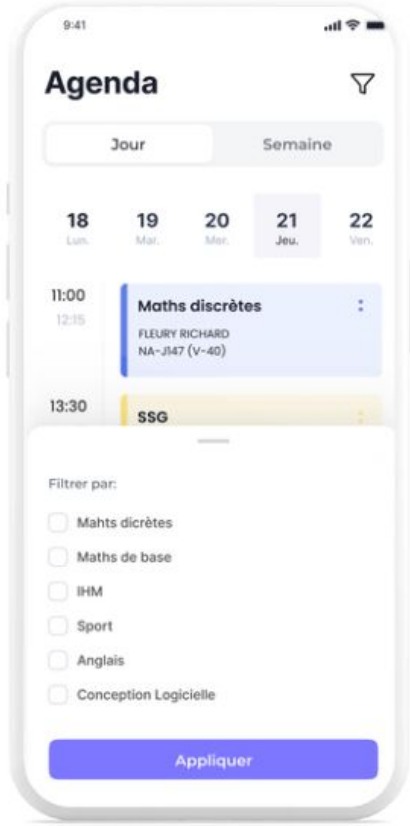
1. Opens on today

The next class sits right under the red "now" line.



2. One tap for details

Room, teacher, type of class and your notes.



3. Make it yours

Choose which courses to show.

Did it work?

71 /100

Usability score (SUS)

From **32** testers. The average for apps is about 68. Old app: **43**.

1< seconds

To find the next class

Our goal was under 10 seconds. Old app: **3 seconds**.

> *"The interface is clear, and the notes system saves lives during rendering."*
>
> **— STUDENT, TESTER V3**

RETROSPECTIVE

Key lessons learned

Critical listening

Do not implement all suggested ideas

Flexibility

Add more clickable navigation trails across the app

Iteration

Detect anomalies through early tests

Observe

Analyze actions rather than listen to statements

Cognitive load

Delete text to lighten the interface

Inclusion

Makes the application simpler for all users

PROJECT N°2

PictoComm App



ABOUT ME

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UX/UI Designer with a Software Engineering Mindset

Engineering student turned designer. I sketch on paper, test, then build the front-end myself.

- **Available for internships**

How I work

The process I followed to design and build PictoComm, an AAC communication app tailored for autistic users and caregivers.

1 Research

Sensory Needs

Benchmark of existing AAC tools & Physical Boards

Cognitive accessibility research

Dual-user personas (child & educator)

2 Plan

Turning needs into a clear structure

User flows

Persistent navigation paths

3 Make

Designing it, then building it

Dark UI prototypes

Java FX/FXML

4 Test

Audit

WCAG 2.2 compliance check

Cognitive load review

ABOUT THE PROJECT

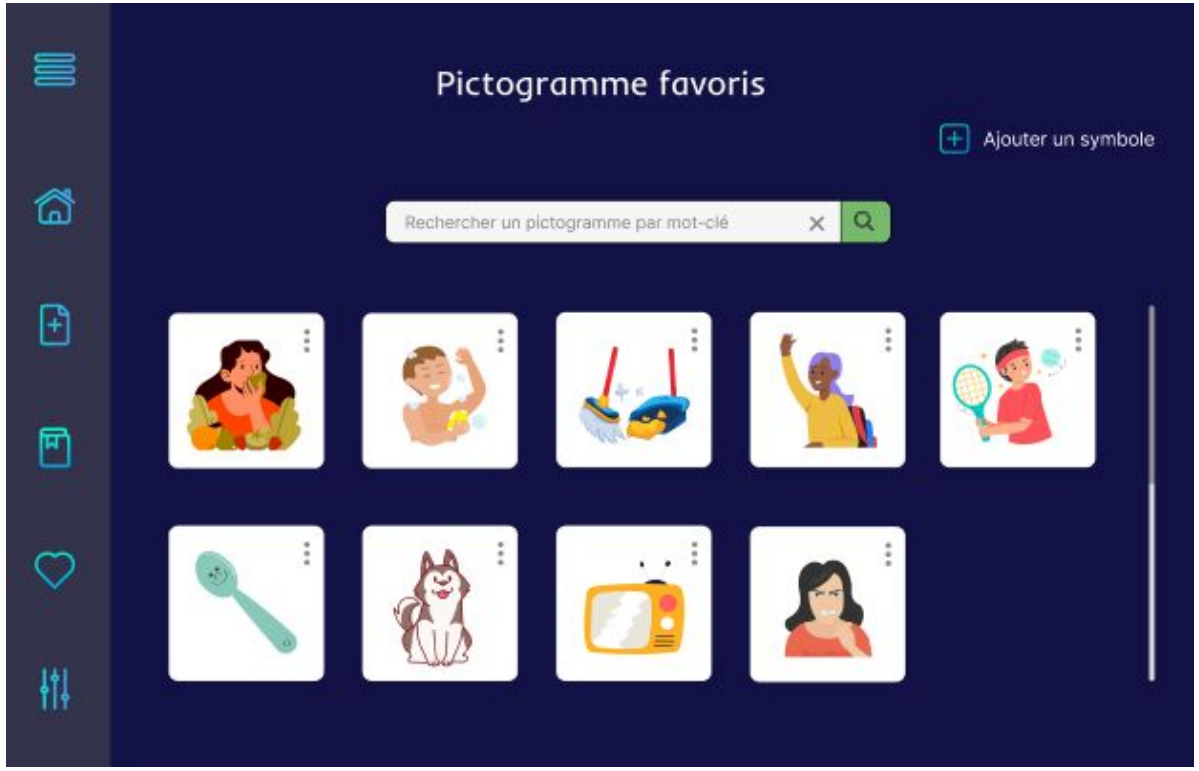
Communication built for specific needs

TIMELINE
3 days, 2024

TEAM
4 students

MY ROLE
Research, UI design, front-end

TOOLS
Figma, ChatGPT, Java FX



WHO IT'S FOR

Autistic learners & caregivers (educators, therapists, parents)

THE PROBLEM

Existing tools cause sensory fatigue, while preparing physical boards takes hours.

WHAT WE BUILT

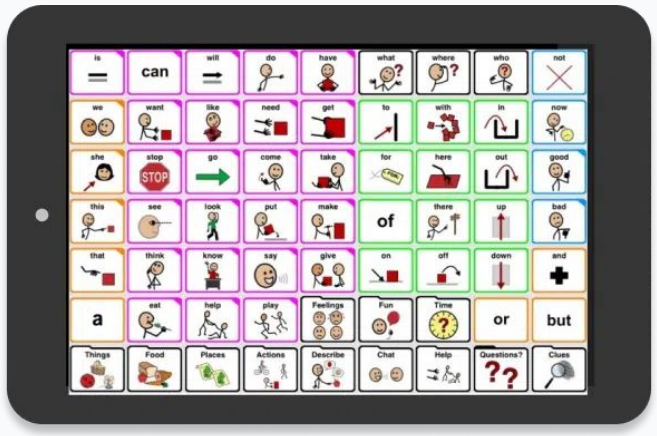
Drag-and-drop routine builder.
WIth WCAG AAA contrast & custom skin tones.

THE PROBLEM

Why standard digital tools fail autistic users

Commercial AAC Apps

Extremely dense grids and steep pricing (200–300€ in a single purchase). Complex nested menus overwhelm autistic users with cognitive friction.



Source: Proloquo2Go App

Generic Routine Apps

Hyper-saturated primary colors, infantilizing animations and zero options to adapt skin tones, borders or text positions.



Source: Avaz AAC App

Physical Boards

Highly tactile and stable but preparation takes hours (printing, cutting, laminating) with constant risk of lost cards.



Translating clinical findings into UI axioms

1. Spatial Predictability

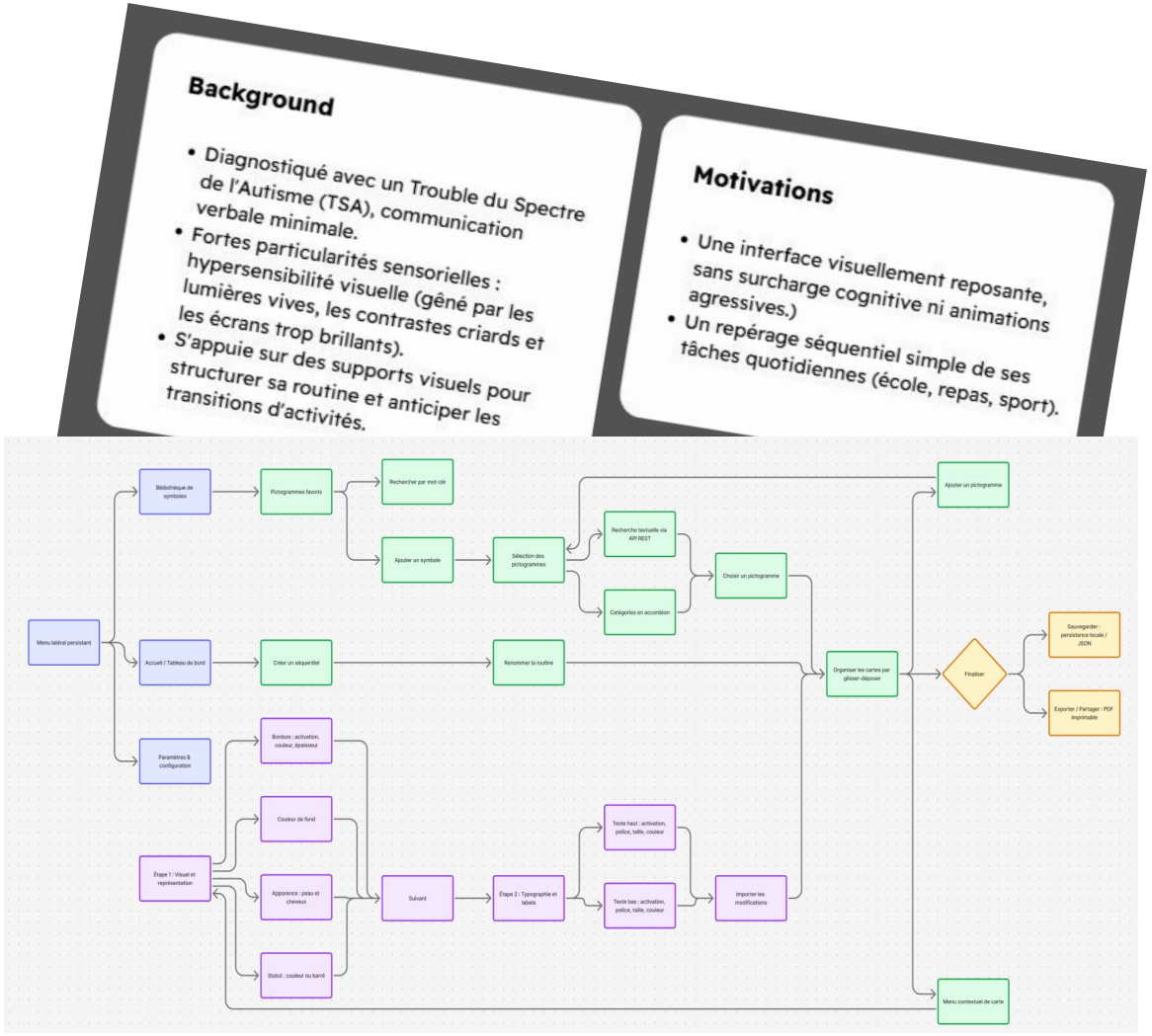
Persistent lateral icons; components must never shift position unannounced.

2. Sensory Soothing

A low-luminance deep navy theme to prevent visual hyper-reactivity.

3. Granular Fine-Tuning

Adjustable border weights, contrast tiers and skin/hair representations.



From research to first sketch: part of the persona and the user flows.

A sensory-friendly dark canvas

1 High Visual Rest

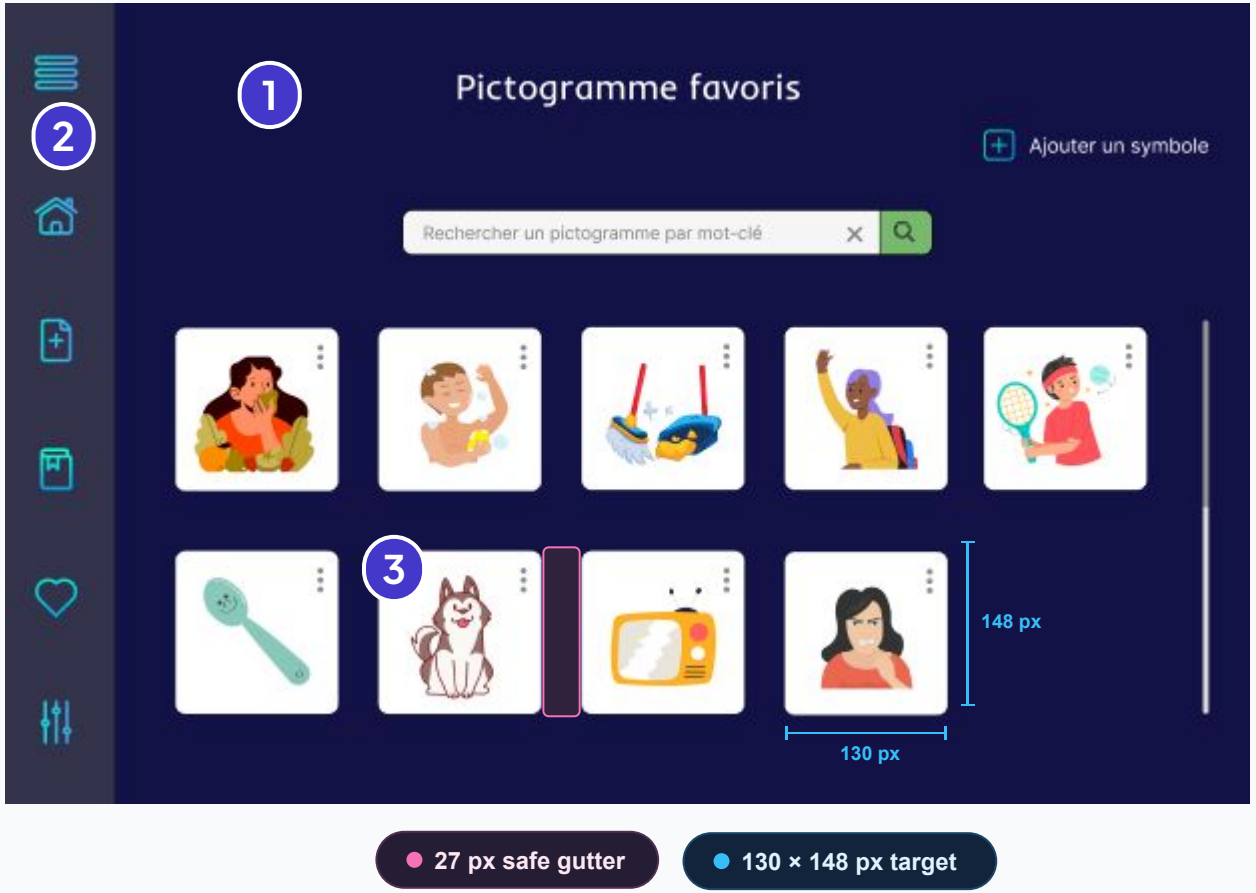
Deep midnight blue (#0d1326) eliminates eye strain and screen glare.

2 Persistent Anchor Bar

Side menu icons maintain stable spatial position to avoid confusion.

3 Touch-Sized Targets

Generous card tiles designed for finger taps (exceeds WCAG AAA 48 px) and safe spacing (exceeds WCAG 2.2 24 px)



Favourite view in the final prototype.

Figma to JavaFX / FXML architecture

1. Design System in Figma

Strict grid, 8pt spacing scale, soothing dark UI palette tokens.

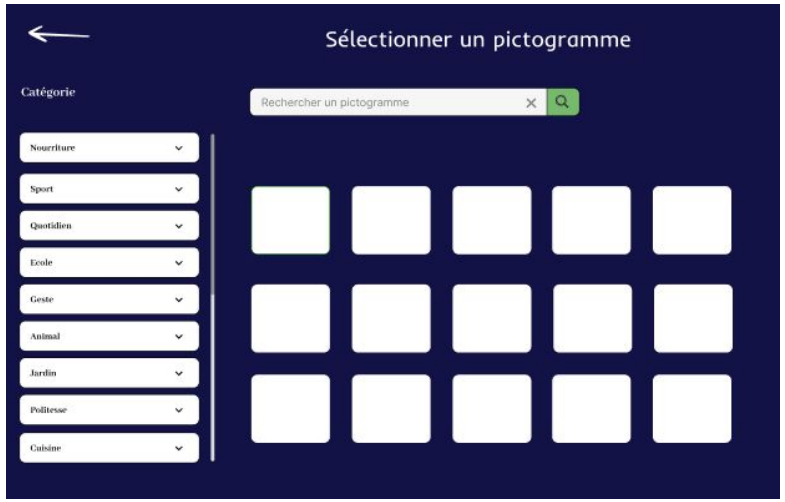
2. JavaFX & FXML Markup

Modular component hierarchy mapping Figma layers directly to FXML.

3. JSON & API Integration

Seamless JSON configuration export and REST API image handling.

Core user journey in 3 steps



1. Search & Pick

Select categorized pictograms or type keyword via API search.



2. Drag into Sequence

Drop tiles in order: breakfast, shower, brushing teeth, clothes.



3. Customize

Adjust border thickness, character skin tone and labels.

Evaluating accessibility & cognitive load

WCAG AAA

Contrast Compliance

Every interactive label and text element passes 4.5:1 ratio minimum.

0

Sensory Clutter

Eliminated animated distractions, popups and aggressive alerts.

100%

Keyboard Focus

Guaranteed sequential keyboard navigation across all interactive buttons.

RETROSPECTIVE

Key lessons learned

MINDSET

Inclusivity benefits everyone

Designing for cognitive sensitivity creates clearer, more resilient software for all users.

METHOD

Radical simplification

Removing unnecessary decorations protects users from cognitive overload.

RESEARCH

Ethics in accessibility

Scientific research and standards bridge the gap when direct user testing is restricted.

THANK YOU

Let’s talk :)

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